Kyaw Soe Lwin

(+1) 650-609-8498 | kyawsoelwin.vercel.app | kylwin@ucsd.edu | linkedin.com/in/kyaw-soe-lwin | github.com/CapMorningStar

EDUCATION

University of California San Diego *Jun 2028*
B.S. Data Science **GPA: 4.0**
Coursework: Machine Learning, Deep Learning, Computer Vision, Statistical Natural Language Processing

EXPERIENCE

Data Science Alliance **San Diego, CA**
Data Science Volunteer *Sep 2026 – Present*

- Designed and executed an end-to-end data auditing and validation pipeline, cross-referencing multi-year counts against source reports to ensure high data integrity (97.5%+ fidelity).
- Standardized schemas and built geospatial crosswalks across 380+ downtown blocks and neighborhood boundaries to enable spatial panel modeling.
- Preparing datasets for time-series decomposition, spatial hotspot analysis (Getis-Ord Gi*), and predictive modeling.

PROJECTS

Local Expert — Offline PDF QA Engine (RAG Pipeline) *2026*

- Built a fully offline, hand-coded RAG pipeline (no LangChain) that answers questions about your own PDFs and cites the exact source page, covering chunking, embedding, vector search, and prompt assembly end to end.
- Split PDFs into overlapping ~800-character chunks, embedded them locally with sentence-transformers, and indexed them in a local Chroma vector database for private, offline retrieval.

LoRA TinyLlama-1.1B Instruction Fine-Tuning *2026*

- Fine-tuned TinyLlama-1.1B with LoRA, adapting only the attention projection layers (q/k/v/o) to cut trainable parameters by over 95% while keeping the base model frozen.
- Used 4-bit quantization to fit the entire training run on a free Colab T4 GPU, then diagnosed and fixed a KV-cache/gradient-checkpointing conflict that was causing garbled output at inference.

Telco Customer Churn Pipeline & Profit Thresholding *2026*

- Built a leakage-audited preprocessing pipeline and tuned an XGBoost model via a 30-trial Optuna search, reaching 0.844 ROC-AUC and 0.671 PR-AUC on a held-out test set of 1,057 customers.
- Built a Streamlit dashboard with SHAP explainability and a profit-threshold analysis that weighs outreach cost against customer lifetime value to recommend an action cutoff.

Real-Time Facial Emotion Detection Engine 2026

- Built a real-time webcam pipeline using OpenCV face detection and a pretrained mini-XCEPTION model (FER-2013, ~66% accuracy) to classify emotions live.
- Rendered live bounding boxes, emotion labels, and a probability bar chart across all 7 emotion categories with low-latency OpenCV rendering.

HONORS

Jack Kent Cooke Undergraduate Scholarship **National**
National Semifinalist *2026*

- Named a National Semifinalist for the Jack Kent Cooke Undergraduate Scholarship, recognizing top-tier high-achieving undergraduates nationwide.

SKILLS

Languages: Python, SQL, Java, TypeScript, Bash/Linux
Frameworks: PyTorch, TensorFlow, Keras, Scikit-Learn, XGBoost, Hugging Face Transformers
Libraries: Pandas, NumPy, OpenCV, ChromaDB, Optuna, SHAP
Tools: AWS, GCP (Vertex AI), Docker, Kubernetes (GKE), PostgreSQL, Git/GitHub, Streamlit, Claude Code
Concepts: Generative AI, LLMs, RAG, PEFT/LoRA, RLHF, Prompt Engineering, Computer Vision, CNNs, Agile/Scrum, OOP